

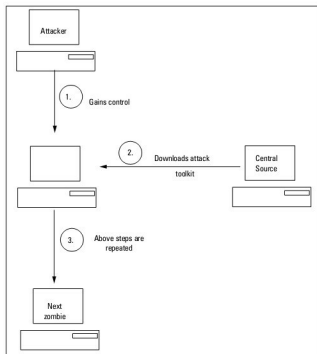


Figure 3: Central source propagation method

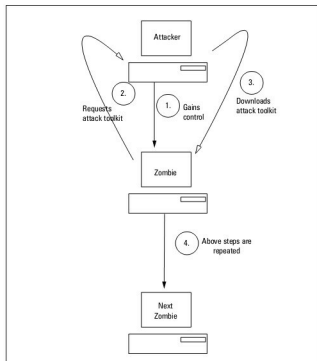


Figure 4: Back-chaining propagation

ICMP Smurf attacks

The Internet Control Message Protocol (ICMP) is used to handle errors and exchange control messages. ICMP can be used to determine if a machine on the Internet is responding. To do this, an ICMP echo request packet is sent to a host. If a host receives the packet, that host will return an ICMP

echo reply packet. A common implementation of this process is the 'ping' application.

In this attack, spoofed IP packets containing ICMP echo requests, with a source address like that of the target system, and a broadcast destination address, are sent to the intermediate network. (Broadcast addresses are specially allocated addresses within all network subnets, used to broadcast messages to the whole network. All hosts within a given subnet receive packets sent to these broadcast addresses and in some cases —ICMP protocol, for instance— respond to them.)

Sending the ICMP echo request to a broadcast address triggers all hosts in the network to respond with an ICMP response packet to the spoofed address (the victim's), thus creating a large mass of packets which are routed to the victim's address. Networks may include up to hundreds of hosts; hence, one attack echo request results in hundreds of packets flooding the victim's site.

UDP flooding attacks

By patching or redesigning the implementation of TCP and ICMP protocols, current networks and systems have incorporated new security features to prevent TCP and ICMP attacks. Nevertheless, attackers may simply send a large amount of UDP packets towards a victim. Since an intermediate network can deliver higher volumes of traffic than the victim network can handle, the flooding traffic can exhaust the victim's connection resources.

Pure flooding can be done with any type of packets. Attackers can also choose to flood service requests so that the victim cannot handle all requests with its constrained resources (i.e., service memory or CPU cycles). UDP flooding is similar to 'flash crowds' that occur when a large number of users try to access the same server simultaneously.

Distributed Reflected Denial of Service (DRDoS) attacks

Unlike typical DDoS attacks, in the case of DRDoS attacks, the army of the attacker consists of master zombies, slave zombies, and *reflectors*. This resembles typical DDoS attacks up to a particular point. The attackers have control over master zombies, which, in turn, have control over slave zombies. The difference in this type of attack is that slave zombies are led by master zombies to send a stream of packets with the victim's IP address as the source IP address, to other, uninfected machines (known as reflectors), causing these machines to connect to the victim. Thus, the reflectors send the victim a greater volume of traffic, because they see the victim as requesting it. Therefore, in DRDoS attacks, the attack is mounted by non-compromised machines, which mount the attack without being aware of the hostile intent.

Comparing the two scenarios of DDoS attacks, we should note that a DRDoS attack is more detrimental than